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FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA

CMH 101 — Past Questions & Materials

by Renance

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Paper 1 of 1 — CMH101 handout 2

Level 100 • First Semester • Handout

Module 1

Atom and Chemical

Reaction

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Unit 1

Atom

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1.0 Introduction

The atom is the basic unit of matter and serves as our starting point for the study of chemistry. The atom is composed of the subatomic particles protons, neutrons, and electrons. Scientists have studied atoms for hundreds of years and have developed a number of different models to describe them, as experimental technology has improved and new discoveries have been made. Models of atoms were postulated and discarded before the right one was discovered. In this module you will be exposed to the various atomic models that are proposed and discarded until the final one was accepted.

2.0 Learning Outcomes

After studying this module, you should be able to:

1. List out the various scientists who attempted to explain Atom.
2. State the name of scientist that discovered neutron
3. Write with at least 70% accuracy, brief accounts of the attempts made by the two scientists and the result of these attempts.
4. Explain William Thomson's Atomic Model
5. State the limitations of Thomson's atomic model

6. Explain the atomic model of the atom proposed by Ernest Rutherford?

3.0 Learning Content

3.1 Attempt made by Democritus

Democritus was a Greek philosopher who was the first person to use the term atom (atomos: meaning indivisible). He thought that if you take a piece of matter and divide it and continue to divide it you will eventually come to a point where you could not divide it any more. This fundamental or basic unit was what Democritus called an atom.

He called this the theory of the universe: All matter consists of atoms, which are bits of matter too small to be seen.

He assumes that:

1. There is an empty space between atoms
2. Atoms are completely solid
3. Atoms have no internal structure
4. Each atom (of a different substance) is different in size, weight and shape.

Interestingly, these thought about atoms are philosophical concepts, but apparently never considered performing experiments to test their ideas.

Self-Assessment Exercises 1

1. Define atom?
2. What is the major contribution of Democritus to the development of atom ?

3.2 Attempt made by John Dalton

John Dalton was the first to adapt Democritus' theory into the first modern atomic model.

He assumes that:

1. An atom is the smallest unit of an element that can participate in a chemical change.
2. Atoms can neither be created nor be destroyed, but are instead rearranged to yield substances that are different from those present before the change
3. Elements are characterized by the weight of their atoms
4. When elements react, it is their atoms that have combined to form new compounds.
5. The atoms of an element are identical in all respects, i.e. size, mass, density, chemical properties, but they differ from the atoms of other elements.

Self-Assessment Exercises 2

1. Who was the first scientist that first to adapt Democritus' theory into the first modern atomic model?
2. What are the major findings of John Dalton in the development of modern atomic model?

3.3 Attempt made by J.J Thomson and Robert Millikan

Joseph John Thomson was a British physicist and Nobel Laureate in Physics, credited with the discovery of the electron, the first subatomic particle to be discovered. He used his research on cathode ray tube (Figure1) technology in this discovery.

Figure 1: Cathode Ray Tube

Deflection of Cathode Rays by an Electric Field. As the cathode rays travel toward the right,

they are deflected toward the positive electrode (+), demonstrating that they are negatively charged.

How does this Tube Work?

1. Nearly Empty tube (Air has been sucked out)
2. An electric charge is passed through the tube. Travels from cathode to anode
3. The charge is invisible, so to see where it traveled a fluorescent screen is placed at back of tube. Where the beam hits, a dot will appear on the screen. You could also use a fluorescent gas and the whole tube will light up.
4. This beam will always travel straight if not interfered with.
5. The deflection coils each has a specific charge. One is positive and the other is negative.
6. Thomson showed that the charge would deflect away from the negative coil. He then stated that this charge was thus a negative charge.

J .J Thomson also predict the mass of the negative charge he discovered. He then found out that this charge was 1000 times lighter than a hydrogen atom. He made a bold statement saying that this negative charge must be inside an atom. This negative charge (he called corpuscles) later became known as the electron.

Subsequently, the American scientist Robert Millikan (1868–1953) carried out a series of experiments using electrically charged oil droplets, which allowed him to calculate the charge on a single electron. With this information and Thomson's mass-to-charge ratio, Millikan determined the mass of an electron:

It was at this point that two separate lines of investigation began to converge, both aimed at determining how and why matter emits energy.

Self-Assessment Exercise 3

1. Give the name of the scientist that discovered the electron in an atom.
2. Which scientist calculate the charge on a single electron. .
- 3.4 Thomson's William Atomic Model

Using what he had discovered, Thomson predicted what an atom should look like. These are the key points to Thomson's Atomic Model:

Thomson atomic model was proposed by William Thomson in the year 1900. This model explained the description of an inner structure of the atom theoretically. It was strongly supported by Joseph John Thomson, who had discovered the electron earlier.

Some of the Assumptions of Thomson's Atomic Model

1. Because of its design this model is known as the plum pudding model
2. An atom consists of a positively charged sphere with electrons embedded in it

Postulate

3. An atom as a whole is electrically neutral because the negative and positive charges are equal in magnitude

Figure 2: Thomson's Atomic Model

Limitations of Thomson's atomic model

- i. It failed to explain the stability of an atom because his model of atom failed to explain how a positive charge holds the negatively charged electrons in an atom. Therefore, This theory also failed to account for the position of the nucleus in an atom
- ii. Thomson's model failed to explain the scattering of alpha particles by thin metal foils
- iii. No experimental evidence in its support

Self-Assessment Exercise 4

State the limitations of Thomson's atomic model

3.5 Attempt made by Ernest Rutherford

Ernest Rutherford was not convinced about the model of the atom proposed by Thomson. He thus set up his now famous Gold Foil Experiment.

The atom, as described by Ernest Rutherford, has a tiny, massive core called the nucleus. The nucleus has a positive charge. Electrons are particles with a negative charge. Electrons orbit the nucleus. The empty space between the nucleus and the electrons takes up most of the volume of the atom.

1. He fired alpha particles (positively charged) at a gold foil.
2. He measured the deflection as the particles came out the other side.
3. Most of the particles did not deflect at all. Every now and then a particle would deflect all the way back.
4. He said that there must be a positive centre of the foil. He called this centre the nucleus.

Figure 3: Rutherford's Atomic Model (Planetary Model)

1. The nucleus of the atom is a dense mass of positively charged particles.
2. The electrons orbit the nucleus
3. A problem raised was: Why are the negatively charged particles not attracted by the positively charged nucleus
4. Rutherford stated that the atom was like a mini solar system and that the electrons orbited the nucleus in a wide orbit. That is why it is known as the planetary model.

Self-Assessment Exercise 5

1. Explain the atomic model of the atom proposed by Ernest Rutherford?

3.6 Attempt made by Niels Bohr

Niels Bohr agreed with the planetary model of the atom, but also knew that it had a few limitations. Using his knowledge of energy and quantum physics he was able to perfect Rutherford's model. He was able to answer why the electrons did not collapse into the nucleus.

Figure 4: Bohr's Atomic Model (Rutherford's - Bohr's Model)

1. Electrons orbit the nucleus in orbits that have a set size and energy.
2. The lower the energy of the electron, the lower the orbit.
3. This means that as electrons fill up the orbitals, they will fill the lower energy level first.
4. If that energy level is full (or at capacity), a new energy level will begin.
5. Radiation is when an electron moves from one level to another. Problems with this theory: Electrons do not travel on a specific orbit or path.

Self-Assessment Exercise 6

1. Explain the Bohr's Atomic Model (Rutherford's - Bohr's Model)

3.7 Attempt made by Erwin Schrödinger

Erwin Schrödinger was a revolutionary physicist who used Heisenberg's uncertainty principle to come up with the atomic model that we still use today. Austrian physicist Erwin Schrödinger (1887-1961) developed an "Electron Cloud Model" in 1926. It consisted of a dense nucleus surrounded by a cloud of electrons at various levels in orbitals. Schrödinger and Werner Heisenberg (1901-1976) mathematically determined regions in which electrons would be most likely found. The probability of the finding the electrons in the orbitals are sometimes referred to as "lobes." They used the mathematical equations for the behaviour of waves following the work on waves by Louis de Broglie (1892-1987).

Schrödinger's Atomic Model (Cloud Model)

1. An electron does not travel in an exact orbit
2. We can predict where it will probably be
3. We cannot say for certain where it is, but only where it ought to be.
4. The type of probability orbit is dependent on the energy level described by Bohr

Figure 5: Schrödinger's Atomic Model (Cloud Model)

Self-Assessment Exercise 7

Do the electron travel in an exact orbit?

4.0 Conclusion

In conclusion, the Scientists tradition of recording and system knowledge gained through observations and experiments has enabled us to learn about the discovering of the atoms. Even some of the assumptions that were not correct were used to further understand the structure and composition of atoms.

5.0 Summary

In this unit, we have discussed the various contribution of different scientist to the development of atom. The smallest part of an element is called an atom and each atom (of an element) is different in structure from other atoms (of other elements). An atom can be divided in smaller subatomic particles: Protons, Electrons and Neutrons. The nucleus is the centre of an atom. It contains protons and neutrons. Electrons orbit the nucleus. As we go up the periodic Table, an electron and proton is added. Electrons occupy a certain energy level (of a certain size). Once the energy level is full, a new level begins. Within each of these levels are special types of orbitals. These depend on the energy level. Each orbital can contain two electrons.

6.0 Tutor-Marked Assignments

1. What is an atom?
2. List out the various scientists who attempted to explain Atom.
3. Which scientist that discovered neutron
4. What are the assumptions of William Thomson's Atomic Model
5. State the limitations of Thomson's atomic model

6. Explain the atomic model of the atom proposed by Ernest Rutherford?

7.0 References/Further Reading

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Unit 2

Chemical Reaction

Content

1.0 Introduction

2.0 Learning Outcomes

3.0 Learning Content

3.1 Chemical Reaction

3.2 Evidence of chemical reactions

3.3 Writing Chemical Equation

3.4 Reactants and Products

3.5 Coefficients and Subscripts in a Chemical Reaction

3.6 Balancing a Chemical Equation

3.7 Chemical Symbols

4.0 Conclusion

5.0 Summary

6.0 Tutor Mark Assignment

7.0 References/Further Readings

1.0 Introduction

The language used in chemistry is seen and heard in many disciplines, ranging from medicine to engineering to forensics to art. The language of chemistry includes its own vocabulary as well as its own form of shorthand. Chemical symbols are used to represent atoms and elements. Chemical formulas depict molecules as well as the composition of compounds. Chemical equations provide information about the quality and quantity of the changes associated with chemical reactions. The concept of a chemical reaction dates back about 250 years. It had its origins in early experiments that classified substances as elements and compounds and in theories that explained these processes. Development of the concept of a chemical reaction had a primary role in defining the science of chemistry as it is known today. The concept of chemical reactions involving the combination of elements clearly emerged from his writing, and his approach led others to pursue experimental chemistry as a quantitative science. This unit will lay the foundation for our study of the language of chemistry. The concepts of this foundation include the understanding of chemical reactions.

2.0 Learning Content

At the end of the unit, students should be able to:

1. Explain Chemical reaction
2. Identify the evidence for chemical reactions
3. Derive chemical equations from narrative descriptions of chemical reactions.
4. Write and balance chemical equations in molecular.
5. Identify the reactants and products in any chemical reaction
6. Convert word equations into chemical equations
7. Use the common symbols such as (g), (l), (s), (aq), and $\rightarrow$ appropriately when writing a chemical reaction.
8. Explain the roles of subscripts and coefficients in chemical equations.
9. Balance a chemical equation when given the unbalanced equation.
10. Explain the role of the Law of Conservation of Mass in a chemical reaction

3.0 Learning Content

3.1 Chemical Reaction

A chemical reaction is a process that leads to the transformation of one set of chemical substances to another. Chemical reactions encompass changes that only involve the positions of electrons in the forming and breaking of chemical bonds between atoms, with no change to the nuclei (no change to the elements present), and can often be described by a chemical equation. In a chemical change, new substances are formed. In order for this to occur, the chemical bonds of the substances break, and the atoms that compose them separate and rearrange themselves into new substances with new chemical bonds. When this process occurs, we call it a chemical reaction. A chemical reaction is the process in which one or more substances are changed into one or more new substances.

Self-Assessment Exercise 1

Define Chemical Reaction

3.2 Evidence of Chemical Reactions

To identify a chemical reaction, we look for a chemical change. A chemical change always produces one or more types of matter that differ from the matter present before the change. The formation of rust is a chemical change because rust is a different kind of matter than the iron, oxygen, and water present before the rust formed. The explosion of nitroglycerin is a chemical change because the gases produced are very different kinds of matter from the original substance.

Self-Assessment Exercises 2

Explain chemical change

3.3 Writing Chemical Equation

One or more chemicals can convert into one or more new substances through a chemical reaction. Chemical equations are used to describe chemical reactions. Reactants appear on the left side of chemical equations, yields are indicated by an arrow ($\rightarrow$), and products appear on the right.

Assessment Exercise 3

What are Chemical equations

3.4 Reactants and Products

We must state the components that are present at the start and the end of a chemical reaction in order to describe it. Reactants (r) are the substances present at the beginning of a reaction, while products (p) are the substances present at the final stage of a reaction.

When reactants are placed in a reaction vessel, a reaction can take place that yields products. Reactants, or whatever we have as our beginning ingredients, are the starting materials. The reactants' interactions with one another in the reaction vessel produce the products, which are simply what is produced.

Reactants (starting materials) → Product (Final stage)

Self- Self-Assessment Exercise 4

Differentiate between reactant and product

3.5 Coefficients and Subscripts in a Chemical Reaction

Chemical equations use two different kinds of numbers. The chemical formulas for the reactants and products have subscripts, and coefficients are added in front of the formulas to show many molecules of the material are consumed or created.

Figure: Balancing Equations. You cannot change subscripts in a chemical formula to balance a chemical equation; you can change only the coefficients. Changing subscripts changes the ratios of atoms in the molecule and the resulting chemical properties.

The subscripts are part of the formulas and once the formulas for the reactants and products are determined, the subscripts may not be changed. The coefficients indicate the number of each substance involved in the reaction and may be changed in order to balance the equation. The equation above indicates that one mole of solid copper is reacting with two moles of aqueous silver nitrate to produce one mole of aqueous copper (II) nitrate and two atoms of solid silver.

Self-Assessment Exercise 5

Differentiate between subscripts and coefficients

3.6 Balancing a Chemical Equation

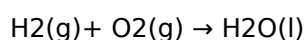
The equation cannot be balanced by altering the subscripts of the reactants or the products since the identities of the reactants and products are fixed. To do so would alter the chemical identity of the species being described.

The process of writing chemical equation are as follows:

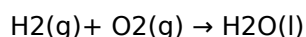
Step 1: Identify the type of reactant and product.

Write a word equation, Hydrogen gas + Oxygen Gas (reactant) → Water (product)

Step 2: Write the chemical formula for each of the reactants and products



Step 3: Count the number of each type of atom in reactants and products

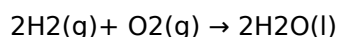


Lift hand side (LHS) : H=2; O=2;

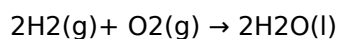
Right hand side (RHS): H =2; O=1

Step 4: Balance the equation:

Compare the number of each type of atom on both sides (reactants and products) and try to balance the reaction by placing appropriate coefficients in front of the reactants and products.



Step 5: Check whether the reaction is balanced or not (atoms on LHS = Atoms on RHS)



LHS: H =4 O=2;

RHS: H =4; O=2

Magnesium reacts with oxygen to produce magnesium oxide. Write the balanced chemical equation for this reaction.

Self-Assessment Exercise 6

Write the balanced chemical equations for the following reactions and identify the type of reaction in each case.

(a) In the thermite reaction, iron (III) oxide reacts with aluminium and gives molten iron and aluminium oxide.

(b) Magnesium ribbon is burnt in an atmosphere of nitrogen gas to form solid magnesium nitride.

(c) Chlorine gas is passed in an aqueous potassium iodide solution to form potassium chloride solution and solid iodine.

(d) Ethanol is burnt in the air to form carbon dioxide and water and releases heat.

3.7 Chemical Symbols

Chemical symbols are the notations that represent chemical elements. Instead of using full names to refer to elements in chemistry, using a symbol made of two letters make things more straight forward. It is a short hand method used to illustrate better chemical reactions in science diagrams based on how we use short names in our everyday lives. Using chemical symbols is all about quickly identifying elements and chemical reactions. Early chemists used to assign short names and strange marks to new elements that developed into specific characters over time. Most of the time, you can quickly identify the element by seeing its symbol, which is the purpose of using them. There are a few special symbols that we need to know in order to "talk" in chemical shorthand. In the Table below is the summary of the major symbols used in chemical equations. Table 1 shows a listing of symbols used in chemical equations.

Table 1: Symbols used in chemical equations

Symbol	Description
(s)	Solid
+	Used to separate multiple reactants of products
(l)	Liquid
Δ	the arrow is used as a catalyst in the reaction
(g)	Gas
→	net forward reaction

(aq) aqueous (dissolved in water) reaction in both
[c]
directions
→ Yield sign; separates reactants equilibrium[d]
from products
replaces the yield sign for stoichiometric relation
reversible reactions that reach =
equilibrium
triangle indicates that the reaction $\leftrightarrow$ resonance (not a
is being heated reaction)
↑ Alternately, an arrow without
↓ parentheses is used in some cases
to indicate formation of a gas ↑ or
precipitate ↓. This is especially
useful if only one such species is
formed.

Self-Assessment Exercise 7

Write and explain four Symbols used in chemical equations

4.0 Conclusion

CONCLUSION

- Chemical reaction - The process in which one or more substances are changed into one or more new substances.
- Reactants - The starting materials in a reaction.
- Products - Materials present at the end of a reaction.
- Balanced chemical equation - A chemical equation in which the number of each type of atom is equal on the two sides of the equation.
- Subscripts - Part of the chemical formulas of the reactants and products that indicate the number of atoms of the preceding element.
- Coefficient - A small whole number that appears in front of a formula in a balanced chemical equation.

5.0 Summary

A chemical reaction is the process by which one or more substances are changed into one or more new substances. Chemical reactions are represented by chemical equations. Chemical equations have reactants on the left, an arrow that is read as "yields", and the products on the right. We use various types of chemical symbols in chemistry. There are many symbols for elements, equations and molecular structures. Using the correct symbols is the only way to illustrate anything related to chemistry like atoms, molecules and reactions. To be useful, chemical equations must always be balanced. Balanced chemical equations have the same number and type of each atom on both sides of the equation. The coefficients in a balanced equation must be the simplest whole number ratio. Mass is always conserved in chemical

reactions.

6.0 Tutor-Marked Assignments

1. Which of the following is a chemical reaction?

- a. Freezing liquid mercury.
- b. Adding yellow to blue to make green.
- c. Cutting a piece of paper into two pieces.
- d. Dropping a sliced orange into a vat of sodium hydroxide.
- e. Filling a balloon with natural air.

- a. Painting a wall blue.
- b. A bicycle rusting.
- c. Ice cream melting.
- d. Scratching a key across a desk.

a.

2. Questions

3. Is each chemical equation balanced?

- a. $2\text{Hg}(\ell) + \text{O}_2(\text{g}) \rightarrow \text{Hg}_2\text{O}_2(\text{s})$ $2\text{Hg}(\ell) + \text{O}_2(\text{g}) \rightarrow \text{Hg}_2\text{O}_2(\text{s})$
- b. $\text{C}_2\text{H}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ $\text{C}_2\text{H}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
- c. $\text{Mg}(\text{NO}_3)_2(\text{s}) + 2\text{Li}(\text{s}) \rightarrow \text{Mg}(\text{s}) + 2\text{LiNO}_3(\text{s})$ $\text{Mg}(\text{NO}_3)_2(\text{s}) + 2\text{Li}(\text{s}) \rightarrow \text{Mg}(\text{s}) + 2\text{LiNO}_3(\text{s})$

7.0 References/Further Reading

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Federal University of Technology Minna

Module 3

Course code CHM111

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CONTENT

Introduction

Learning Outcomes

Bonds

Types of Chemical Bonding

Conclusion

Introduction

■ what

■ how and

- effect of being together of these atoms.

Learning Outcomes

After studying this module, you should be able to:

- state the basis of chemical bonding
- explain the concept ionic bond
- enumerate the types of bonds
- demonstrate how chemical bonding works out ,
- specify bond types of given molecules.

Bonds

- 2 - electron/2 - centre bonds

e e

A _____ B

Scheme 1. 2-electron/2-center bond formation

Atom A and B must give two electrons from two centres (A and B) to form a chemical bond. For example, BeCl_2 , Be is in group II and therefore has two valence electrons, one is used to form a Be-Cl bond and the other used to form a second Be-Cl bond

Types of chemical bonds

- Ionic Bonds
- Covalent Bond
- Metallic Bond
- Van der Waals Bond
- Hydrogen Bond
- Conclusion

- Ionic Bonds

- a metal and a nonmetal
- completely transferred from one atom to another
- attracted to each other by electrostatic forces.
- Bond dissociation energy of about 500kJmol^{-1}
- Hardness and high melting point
- Ions are free in solution and so high electrical conductivity
- Examples:

1. $\text{Na}^+ + \text{Cl}^- \rightarrow \text{NaCl}$ 2. $\text{Ca}^{2+} + \text{Cl}^- \rightarrow \text{CaCl}_2$ 3. $\text{Al}^{3+} + \text{S}^{2-} \rightarrow \text{Al}_2\text{S}_3$

- Covalent Bond

- Covalent bonds usually form between two nonmetals or a metalloid and a nonmetal
- the atoms share one or more pairs of electrons (by using their valence electrons) between each other to obtain a filled valence level
- shared pair of electrons” or “bonding pair of electrons”
- “unshared pair of electrons” or “lone pairs”

■ The structure of a covalent crystal is related to the number of valence electrons, the nature of orbitals involved in bond formation and their orientation. One of the most commonly cited examples is that of diamond .

• Each carbon atom in diamond is tetrahedrally bonded to four neighbouring carbon atoms. These orbitals overlap with the similar set of orbitals on the neighbouring atoms.

• Metallic Bond

■ According to the bond theory, the metallic solids can be considered as having simple covalent bonds between adjacent atoms.

■ However-in these cases, the number of electron pairs available for bond formation is less than the number of orbitals available.

■ In the band theory of metals, a crystalline metallic solid is considered as a single giant molecule. Linear combination of atomic orbitals on all the atoms is taken to give molecular orbitals of the solid just as in the case of simple diatomic molecule. It is also assumed that there is negligible overlap of inner shell atomic orbitals and the energies of these remain practically the same as atomic orbitals on isolated atoms.

• Van der Waals bond

Van der Waals forces include:

i) Dipole-dipole interactions

■ Polar molecules can attract each other electrostatically.

■ During this attraction the positive end of one molecule is close to the negative end of the adjacent molecule.

i) Dipole-induced dipole interactions

■ attractive forces between polar molecules at ordinary temperatures whereas at high temperatures it cannot.

■ A polar molecule can induce a dipole moment in a neighbouring polarisable atom or molecule. Let us explain,

the terms 'polarisable' and 'polarisability'.

An atom or molecule is said to be polarisable. if its electron cloud can be distorted. The ability of a species to

undergo electronic distortion is described in terms of polarisability. Reasons:

i. the electrons are more in number

ii. the influence of the nucleus is less due to larger distance

So a larger atom has a higher polarisability than a smaller atom. For example, argon has higher polarisability than

helium. Similarly applies to molecules. For example, ethane is more polarisable than methane; propane is more

polarisable than ethane, and so on.

i) London or dispersion forces

■ cannot explain the liquefaction of gases like hydrogen, oxygen, chlorine, helium and argon (which are all non polar).

■ London gave an acceptable quantitative explanation for the attractive forces existing

between non polar molecules and hence such forces are called London forces. These forces are called dispersion forces since the oscillations producing the attractive forces are also responsible for the dispersion of light by the molecules.

- Hydrogen Bonding

There are several specific types of interactions encountered between various types of molecules. Of these, metallic bonding and hydrogen bonding are very significant. Here let us study hydrogen bonding in detail. When a hydrogen atom is covalently bonded to a strongly electronegative atom, such as oxygen, fluorine or nitrogen, the bond is much polar.

Such a hydrogen atom would still possess large affinity for nonbonding electrons present on other oxygen, nitrogen or fluorine atom. The latter atom could be a part of the same molecule or a neighbouring molecule. The strong interaction that results is called a hydrogen bond. It is a special type of dipole-dipole attraction. In water, for example, hydrogen bonding arises between hydrogen atom (positive end of the dipole) of one water molecule and the oxygen atom (negative end of the dipole) of the other (Fig.3.1a). Hydrogen fluoride is another molecule having hydrogen bonding (Fig. 3.1b).

Hydrogen bonding is strong in HF, H₂O and NH₃ as compared to many hydrides due to the higher electronegativity of fluorine, oxygen and nitrogen. Strong hydrogen bonding in these compounds results in enhanced attractive interactions between the molecules.

Conclusion

The chemical bonding is essentially a force operating when atoms or molecules are in vicinity of one another. Some lead to strong bond like in metallic and ionic bonds.

Covalent as well as Van der Waals as well as hydrogen bonds are not as strong.